

RG03

Climate of Rajasthan

Rajasthan Geography gives 11-15 questions every paper, and climate reliably contributes 1-3 of them. The favourites are the Koppen type of a named district, the highest and lowest rainfall district, the Phalodi record, and local winds matched to their meaning. Matching and one-liner formats dominate.

This chapter is about one simple fact and its consequences - Rajasthan is dry, and it gets drier as you move north-west. Everything RPSC asks here hangs off that gradient: the 50 cm isohyet, the four Koppen types, the rainfall extremes, the temperature records, the local winds. It is a small chapter with a very high strike rate because the facts are fixed numbers and fixed place names.

Climate of Rajasthan

Basic character

- Arid to semi-arid
- Average rain about 58 cm
- India average about 125 cm
- Arid tract about 61 per cent of state

Rainfall gradient

- Falls south-east to north-west
- 50 cm isohyet on Aravalli axis
- Jhalawar highest district
- Mount Abu wettest station
- Jaisalmer / Sam driest

Koppen types

- Aw - Vagad (Banswara, Dungarpur)
- BWhw - Jaisalmer, Bikaner, Churu
- BShw - Shekhawati, Nagaur, Luni basin
- Cwg - Jaipur, Alwar, Kota, Udaipur

Temperature

- Phalodi 51.0 C, 19 May 2016
- Churu - extreme continental
- Mount Abu coldest
- Fatehpur (Sikar) sub-zero on plains

Winds and systems

- Loo - May-June westerly
- Andhi - pre-monsoon dust storm
- Bhabulya - dust whirlwind
- Mavath - winter rain
- Western disturbances - Mediterranean

Monsoon

- Enters last week of June
- Entry via Banswara-Dungarpur
- Arabian Sea branch dominant
- Withdraws by end September
- About 90 per cent of annual rain

Hazards and policy

- Drought - Jaisalmer, Barmer, Bikaner
- Chhappaniya Akal VS 1956
- Heat waves, cold waves, hailstorms
- SAPCC 2010 under NAPCC 2008

Start here - why Rajasthan is dry at all

Rajasthan sits in the path of the south-west monsoon. So why is half of it desert? Because three things go wrong at once. The monsoon current arrives here already spent, the one mountain range that could have wrung it out is

pointing the wrong way, and the state is a long way from the sea. Get this causal chain right before you memorise any number - RPSC asks it as an assertion-reason or a statement question almost as often as it asks the numbers.

- The Aravalli range runs south-west to north-east - roughly parallel to the monsoon current.
- Parallel means the winds travel along the range, not up it, so there is almost no orographic lifting.
- The Arabian Sea branch that crosses western Rajasthan is not forced to rise and so does not shed its moisture.
- The Bay of Bengal branch reaches eastern Rajasthan only after crossing the whole Gangetic plain, badly weakened.
- Continentality - distance from the sea - plus a bare sandy surface gives extreme heating and very low humidity.

TRAP

Do not write that the Aravallis 'block' the monsoon. They fail to block it. That failure is precisely the point - and the wrong option in the exam is always the one that says the western side gets more rain than the eastern side.

ASKED IN RAS

A standard RPSC framing: Assertion - western Rajasthan is arid though it lies in the monsoon path. Reason - the Aravallis are parallel to the monsoon current and do not obstruct it. Both true, and the reason correctly explains the assertion.

Rainfall – the most examined part of this chapter

Learn rainfall as a gradient, not as a list. Rain falls off steadily from the south-east corner to the north-west corner. Jhalawar at one end, Jaisalmer at the other, and everything in between sits on that line. If you can draw the diagonal, you can answer most rainfall questions without memorising district figures.

- State average annual rainfall – about 58 cm, against India's average of about 125 cm.
- About 90 per cent of it falls in the south-west monsoon season, late June to mid-September.
- The remaining tenth comes mainly from winter western disturbances.
- Only about 10 per cent of the state's area gets more than 80 cm, and that area lies in the south and south-east.
- Rainfall variability exceeds 60 per cent in the west and is lowest in the south-east – which is why drought is a western problem.

The five rainfall regions of Rajasthan

Region	Annual rainfall	Where it lies
Arid	Below 20 cm	Jaisalmer and the extreme western frontier
Semi-arid	20–40 cm	Bikaner, Churu, Barmer, parts of Jodhpur and Nagaur
Sub-humid	40–60 cm	Central belt along and just east of the Aravallis
Humid	60–80 cm	Eastern and south-eastern plain
Very humid	Above 80 cm	Jhalawar, Banswara, the Mount Abu tract

The 50 cm isohyet – one line, many questions

An isohyet is a line joining places of equal rainfall. The 50 cm isohyet is the one that matters in Rajasthan. It runs broadly along the Aravalli axis and splits

the state into two halves - arid western desert on one side, semi-humid eastern plain on the other. When a question asks what the 50 cm line separates, that is the answer: the arid west from the semi-humid east.

ASKED IN RAS

Asked repeatedly - which district of Rajasthan receives the highest average annual rainfall? Answer: Jhalawar. But the single wettest station is Mount Abu in Sirohi, lifted above 1,200 m on the Aravalli crest.

ASKED IN RAS

Its twin - which district receives the lowest rainfall? Answer: Jaisalmer, of the order of 10-20 cm a year. The driest tract within it is around Sam.

Keep district and station apart in your head. The highest-rainfall DISTRICT is Jhalawar. The wettest PLACE or STATION is Mount Abu. RPSC sets both options side by side in the same question and lets you pick the wrong one, so read whether the stem says district or place.

YAAD RAKHO

Yaad rakho the diagonal - JHALAWAR, BANSWARA, KOTA, JAIPUR, NAGAUR, BIKANER, JAISALMER. Read it as one falling line from the south-east to the north-west, and every ordering question is solved.

Koppen classification of Rajasthan - four types, four regions

Koppen classified climates by letters. The first capital letter is the broad group, the small letters describe the rain and heat. Rajasthan has exactly four of his

types. That is a gift - four rows, learn them cold, because the district-to-type match is the single most reliable climate question in the paper.

Koppen types matched to the regions of Rajasthan

Koppen type	What it means	Districts and region
Aw	Tropical humid, savanna type with dry winter	Vagad - Banswara, Dungarpur, Pratapgarh and the adjoining south-east
BWhw	Hot desert, dry winter	Jaisalmer, Bikaner, Churu, Sri Ganganagar, Hanumangarh, western Barmer, western Jodhpur
BShw	Semi-arid steppe, hot, dry winter	Luni basin, Nagaur, Shekhawati (Sikar-Jhunjhunu), Ghaggar plain, Jalore, Pali
Cwg	Sub-humid, dry-winter monsoon type	Jaipur, Alwar, Bharatpur, Ajmer, Bhilwara, Bundi, Chittorgarh, Udaipur, Kota

- B is the dry group. BW is true desert, BS is steppe - W for wüste (desert), S for steppe.
- The small h stands for hot, the small w for a dry winter, the g for the Gangetic type of temperature regime.
- The order Aw - Cwg - BShw - BWhw is also the order of falling rainfall, south-east to north-west.

- Trick to remember: A for the Adivasi belt - Aw = Banswara and Dungarpur.
- C for Capital - Cwg = Jaipur and the eastern plain.
- BS for the Shekhawati Steppe; BW for the Western desert.
- The second character of the code tells you the region every time.

TRAP

Two traps here. First - Bikaner is BWhw (hot desert), NOT BShw. Second - Am, the tropical monsoon type with a short dry season, does NOT occur in Rajasthan. Any option offering Am is the wrong one.

Temperature – the records, the ranges and the cold

Rajasthan holds India's hottest record and also produces sub-zero winter nights on the plains. Both come from the same cause. A bare sandy surface with no moisture and no cloud heats furiously by day and radiates all of it away by night. That gives the state among the highest diurnal and annual temperature ranges in India.

Temperature records you must be able to quote

Record	Place	Detail
Highest air temperature ever recorded in India	Phalodi	51.0 degrees C on 19 May 2016
Extreme continental station	Churu	Among the highest annual temperature ranges in India; crosses 50 C in summer, falls below freezing in winter
Coldest place in Rajasthan (normally)	Mount Abu, Sirohi	The state's only hill station
Coldest plains station	Fatehpur, Sikar	Most frequent sub-zero minima on the plains, with Churu

ASKED IN RAS

Asked directly – the highest air temperature ever recorded in India, 51.0 degrees C on 19 May 2016, was registered at Phalodi. Note the date; RPSC has used the year as a distractor.

- Phalodi was then a tehsil of Jodhpur district; since the 2023 reorganisation it is a district in its own right and was retained in the December 2024 review.
- Summer maxima above 45 C are routine across Barmer, Jaisalmer, Bikaner and Churu in May and June.

- The three causes of the extreme range - continentality, bare sandy surface, clear cloudless skies with low humidity.

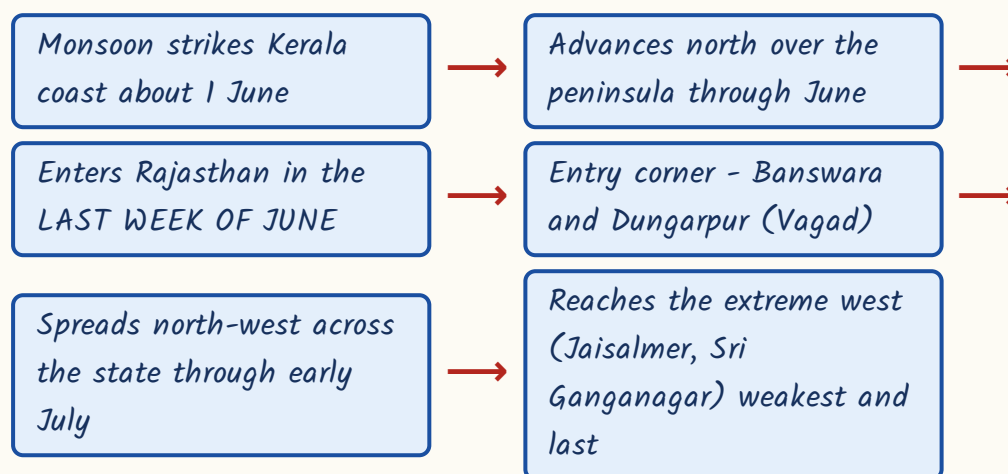
TRAP

Coldest PLACE in Rajasthan is Mount Abu. But if the question says 'on the plains' or 'excluding hill stations', the answer flips to Fatehpur (Sikar) or Churu. Read the qualifier.

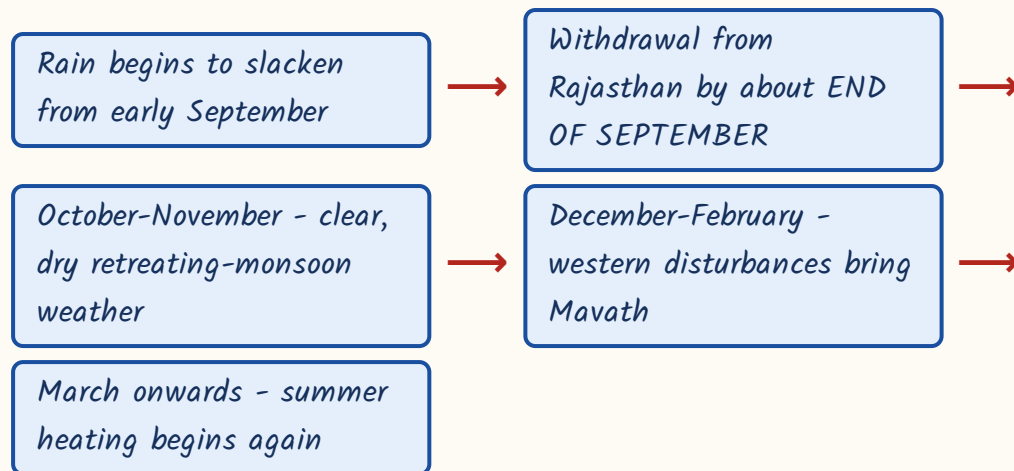
The monsoon over Rajasthan - onset to withdrawal

The monsoon reaches Rajasthan last and leaves it first. It enters through the humid south-east corner in the last week of June and is gone by about the end of September. Roughly three months of rain, and nine months of nothing but western disturbances. Learn the entry corner and the exit month - they are the two asked facts.

Onset of the south-west monsoon over Rajasthan



Withdrawal and the dry months



The two monsoon branches over Rajasthan

Arabian Sea branch	Bay of Bengal branch
Supplies most of western and southern Rajasthan	Supplies eastern Rajasthan
Travels roughly parallel to the Aravallis, so little lifting	Arrives after crossing the entire Gangetic plain
Moisture retained but not released - the cause of western aridity	Already much weakened and depleted by the time it arrives
Dominant branch for the state as a whole	Secondary branch, eastern districts only

- About 90 per cent of Rajasthan's annual rainfall comes in this single season.
- The Tropic of Cancer clips the extreme southern edge of Banswara district - almost the whole state is sub-tropical, not tropical.

Local winds and winter rain - a guaranteed matching question

Four local terms carry almost all the marks in this section. Loo and Andhi belong to the summer, Bhabulya to the arid west at any hot hour, and Mavath to winter. RPSC's favourite trick is to shift one of them into the wrong season.

Local winds and weather terms of Rajasthan

Term	What it is	Season and area
Loo	Hot, dry, dust-laden westerly wind; a major cause of heat-stroke deaths	May-June, western Rajasthan - Barmer, Jaisalmer, Bikaner
Andhi	Violent pre-monsoon dust storm that cuts visibility sharply	May-June, often just before the first monsoon showers
Bhabulya	Small, short-lived local dust whirlwind	Bikaner and the adjoining western tract
Mavath	Light winter rain brought by western disturbances	December-February, chiefly north and north-west Rajasthan

Mavath deserves a sentence of its own. It is only about a tenth of the year's rain, but it falls exactly when wheat, mustard and gram need it. That makes it disproportionately important to the rabi crop and to the state's food economy. Western disturbances themselves are extra-tropical systems - they form over the Mediterranean Sea, travel eastwards, and enter India from the north-west.

TRAP

Mavath is caused by WESTERN DISTURBANCES, not by the retreating or north-east monsoon. That single substitution is the standard 'NOT correctly matched' option.

- Sort them by season and the match writes itself - Loo and Andhi are hot, so summer.
- Mavath is meetha for the farmer - winter rain, rabi crop.
- Bhabulya is a bhanwar, a whirl - a local dust devil, any hot afternoon in the west.

Hazards, droughts and the great famines

Rajasthan's climate hazards follow the rainfall gradient too. Where rainfall is low and variability is high, drought is chronic. Where the surface is bare and hot, heat waves kill. Cold waves come with northerly incursions in winter, and hailstorms ride in on western disturbances. Note that last one carefully - hail is a winter and pre-monsoon phenomenon here, not a monsoon one.

- Drought frequency is highest in Jaisalmer, Barmer and Bikaner - the low-rain, high-variability west.
- High inter-annual rainfall variability, above 60 per cent in the west, is the principal cause of recurrent drought.
- Heat waves peak in May and June in the western districts.
- Cold waves follow the incursion of cold northerly winds in December and January.
- Hailstorms come with western disturbances in winter and the pre-monsoon months and are a major threat to the rabi crop.

The famine you must know by name is the Chhappaniya Akal - literally 'the famine of fifty-six', named for Vikram Samvat 1956, which corresponds to 1899-1900 CE. It is the most severe famine in Rajasthan's recorded history and passed into folk memory and folk song. The earlier Trikal famine of Vikram Samvat 1868 is the other landmark famine name.

- Chhappaniya Akal - VS 1956 = 1899-1900 CE. Convert by subtracting 57 from the Samvat year.
- Trikal - VS 1868, the earlier great famine.
- The Thar or Great Indian Desert is India's largest desert; the arid western tract covers about 61 per cent of Rajasthan's area.
- Eastward expansion of the arid tract is the long-run climate worry - it is why shelterbelts and the Aravalli Green Wall exist.

One last caution on drought. Jhalawar and Banswara are the wettest districts of the state. They can never be the answer to a most-drought-prone question, however plausible the option is made to look. Drought in Rajasthan is always a western answer - Jaisalmer, Barmer, Bikaner.

Climate policy and the current file

The policy layer is thin but it is asked, usually as a one-liner about which national framework the state plan sits under. Every Indian state prepares a State Action Plan on Climate Change consistent with the National Action Plan on Climate Change of 2008 and its national missions. Rajasthan's first such plan was prepared in 2010.

- National Action Plan on Climate Change - 2008, the umbrella framework with its national missions.
- Rajasthan State Action Plan on Climate Change - first prepared in 2010, under that umbrella.
- Rajasthan also has a Climate Change Policy dated 2023 and district-level Heat Action Plans issued by the Disaster Management and Relief Department.

CURRENT LINK

Current link (as of September 2026): heat is now a policy subject in Rajasthan, not just a weather fact. The state's Disaster Management and Relief Department issues district Heat Wave Action Plans each summer, and the pressure to treat heat waves as a notified disaster has been a live national debate since the 2024 heat season. Expect a current-affairs question phrased around heat action plans rather than around temperature records.

REVISION RECAP

1. Rajasthan average rainfall about 58 cm; India's about 125 cm; about 90 per cent falls in the south-west monsoon.
2. Rain falls off from the south-east to the north-west - the whole chapter is that one diagonal.
3. 50 cm isohyet runs along the Aravalli axis and divides the arid west from the semi-humid east.
4. Five rainfall regions: arid below 20, semi-arid 20-40, sub-humid 40-60, humid 60-80, very humid above 80 cm.
5. Highest rainfall district = Jhalawar. Wettest station = Mount Abu. Lowest = Jaisalmer, driest tract Sam.
6. Only about 10 per cent of the state's area gets over 80 cm; variability tops 60 per cent in the west.
7. Koppen: Aw = Banswara-Dungarpur; BWhw = Jaisalmer-Bikaner-Churu; BShw = Shekhawati-Nagaur-Luni; Cwg = Jaipur-Kota-Udaipur.
8. Am does not occur in Rajasthan; Bikaner is BWhw, not BShw.
9. Phalodi 51.0 degrees C on 19 May 2016 - India's highest ever. Churu is the extreme continental station.
10. Mount Abu coldest overall; Fatehpur (Sikar) and Churu coldest on the plains, often sub-zero.
11. Monsoon enters the last week of June through Banswara-Dungarpur and withdraws by about end September.
12. Arabian Sea branch feeds west and south; Bay of Bengal branch, weakened, feeds the east.
13. Aravallis lie parallel to the monsoon current - no orographic lifting, hence western aridity.
14. Loo and Andhi = May-June; Bhabulya = dust whirlwind of the Bikaner tract; Mavath = winter rain from western disturbances.
15. Western disturbances originate over the Mediterranean and enter India from the north-west; they give only about a tenth of the rain.
16. Hailstorms ride on western disturbances in winter and pre-monsoon - not on the monsoon.
17. Drought is worst in Jaisalmer, Barmer, Bikaner; Chhappaniya Akal = VS 1956 = 1899-1900 CE; Trikal = VS 1868.
18. Thar is India's largest desert; the arid tract is about 61 per cent of Rajasthan's area.
19. Tropic of Cancer clips southern Banswara - nearly the whole state is sub-tropical.

20. *Rajasthan SAPCC first prepared 2010, under the National Action Plan on Climate Change, 2008.*